

HINTERRHEIN, SwitzerlandWMO: **067881**Lat: **46.524N**Lon: **9.184E**Elev: **1618**StdP: **83.34**Time Zone: **1.00 (EUC)**Period: **01-11**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-19.3	-17.1	-25.2	0.5	-14.0	-22.8	0.6	-15.0	15.1	-0.7	11.1	-1.2	0.9	230	0.722

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.7	22.7	13.8	21.2	13.4	19.8	13.0	15.2	20.4	14.4	19.4	13.6	18.4	3.7	60

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
13.0	11.4	17.8	12.2	10.8	16.8	11.5	10.3	15.9	48.5	20.5	45.8	19.4	43.4	18.5	17.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 10.8	(b) 8.9	(c) 7.5	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(n) N/A	(o) N/A	(p) N/A		
			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	3.3	-6.7	-5.6	-2.0	3.0	7.2	11.2	12.7	12.1	8.4	5.2	-0.5	-6.0
	DBStd	7.94	4.66	4.21	4.31	3.43	3.15	3.28	2.78	2.78	3.09	3.69	4.62	5.05
	HDD10.0	2707	518	437	372	209	98	26	9	12	64	152	314	494
	HDD18.3	5488	777	671	630	459	347	215	174	193	297	409	564	753
	CDD10.0	259	0	0	0	0	9	61	93	78	17	2	0	0
	CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH23.3	21	0	0	0	0	0	3	3	14	0	0	0	0
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	3.0	2.8	3.0	3.4	3.4	3.2	2.7	2.7	2.8	3.0	3.5	3.1	2.9
Precipitation	PrecAvg	1321	68	63	69	100	125	136	135	151	130	127	142	75
	PrecMax	1796	157	185	175	288	361	271	228	224	318	435	581	161
	PrecMin	827	18	14	9	15	39	38	64	24	26	15	25	5
	PrecStd	239	39	45	43	62	74	66	43	46	86	85	121	40
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.1	6.3	10.0	16.5	19.7	23.9	23.8	25.3	20.4	17.3	11.9	5.1
		MCWB	0.8	1.3	4.0	7.6	11.3	13.9	14.3	13.5	12.5	9.2	6.6	1.3
	2%	DB	3.2	3.8	7.4	13.7	17.3	22.2	22.3	22.5	18.0	14.5	9.1	3.6
		MCWB	0.3	0.3	2.7	6.7	9.6	13.5	14.0	13.8	11.1	7.8	5.2	1.2
	5%	DB	1.7	2.1	5.9	11.1	15.7	20.7	21.0	20.5	16.2	12.7	6.8	2.6
		MCWB	-0.4	-0.5	1.9	5.3	9.0	13.0	13.8	13.2	10.4	7.6	3.6	0.6
	10%	DB	0.4	0.7	4.4	8.9	13.4	18.8	19.6	18.4	14.3	11.2	5.3	1.2
		MCWB	-1.4	-1.8	1.2	4.2	8.3	12.7	13.3	12.8	9.9	7.8	3.2	-0.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.6	2.2	4.5	8.6	12.1	16.0	16.2	16.1	13.8	10.5	8.0	3.0
		MCDB	4.5	5.1	8.3	14.5	18.4	21.0	21.0	21.9	18.8	14.0	10.7	4.1
	2%	WB	0.8	0.8	3.3	7.0	10.8	14.7	15.3	14.7	12.3	9.7	5.5	1.9
		MCDB	2.2	3.2	6.2	12.7	15.9	20.3	20.4	19.5	16.1	12.6	8.3	3.1
	5%	WB	-0.2	-0.4	2.5	5.9	9.7	13.7	14.5	13.9	11.3	8.9	4.3	0.7
		MCDB	1.5	1.7	5.2	10.2	14.3	19.1	19.4	18.7	14.4	11.6	6.2	2.1
	10%	WB	-1.4	-1.3	1.6	4.7	8.6	12.7	13.7	13.1	10.4	7.8	3.4	-0.3
		MCDB	0.4	0.5	4.1	8.3	12.5	18.0	18.4	17.6	13.5	10.5	5.1	1.2
Mean Daily Temperature Range	5% DB	MDBR	8.9	9.9	9.4	8.4	9.0	10.3	10.7	9.7	9.2	8.7	7.1	7.7
		MCDBR	9.4	11.1	11.1	12.7	13.5	14.9	13.8	14.8	13.6	13.0	9.5	7.7
		MCWBR	7.0	8.4	7.2	6.7	7.1	7.4	6.8	7.2	7.4	7.1	6.0	5.6
	5% WB	MCDBR	8.4	9.4	9.3	11.7	11.3	13.1	12.3	12.4	10.9	9.2	7.9	6.1
		MCWBR	6.4	7.2	6.1	6.4	6.3	6.9	6.6	6.6	6.1	5.6	5.4	4.9
Clear-Sky Solar Irradiance	taub	0.237	0.261	0.303	0.348	0.363	0.372	0.367	0.359	0.338	0.312	0.269	0.235	
	taud	2.331	2.267	2.206	2.197	2.295	2.315	2.330	2.369	2.415	2.463	2.450	2.366	
	Ebn at Noon	880	919	920	905	902	892	892	886	876	851	834	844	
	Edh at Noon	72	96	120	134	127	125	121	112	98	80	65	62	
All-Sky Solar Radiation	RadAvg	1.60	2.58	3.79	4.46	4.77	5.62	5.61	4.87	3.79	2.52	1.56	1.34	
	RadStd	0.20	0.30	0.55	0.73	0.48	0.50	0.49	0.43	0.37	0.30	0.23	0.17	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (120 neighbors)	N/A	N/A	N/A	N/A	+0.72	+0.41	+0.31	N/A	N/A	+59	+16

Nomenclature: See separate page